

Statistical Modeling

David Puelz

The University of Austin



- Use supervised learning to build models for prediction and for understanding (causal inference)
- Understand the benefits and limitations of the models we build
- Given a new situation, select an appropriate model, carry it out, and effectively communicate the results
- This course focuses on practice and implementation. But ... we will pick up theoretical details when necessary.



- **Educator:** Teaching at UATX since 2024, UT Austin from 2020-2024, graduate student from 2013-2018. Previously at the University of Chicago Booth School of Business and Goldman Sachs.
- **Researcher:** I develop new statistical/machine learning methods for analyzing complex data, with applications in policy, economics, and the social sciences.
- **Human:** Grew up in Dallas. Husband and dad with 2 girls + 1 boy + 1 dog. I like working out, smoking meat, golf, and cycling.



Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff

Answer two questions for me:

What is one thing you're *excited* about for this class?

What is one thing you're *nervous or concerned* about for this class?

Pass around your answers. Another student will read your responses, so everyone stays anonymous.

Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff



- Access at uaustin.populiweb.com
- Access at <https://github.com/dpuelz/StatisticalModeling>
- Github pages will contain **schedule**, **homeworks**, **data**, **code**, ...
- **Github will be our main landing page.** Populi will be used for grades and administrative tasks.

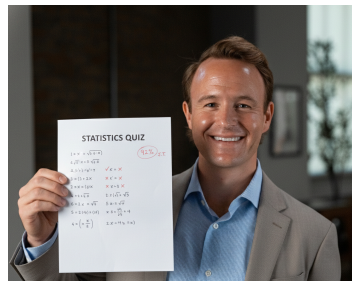


- Understanding the concepts really only comes from practice

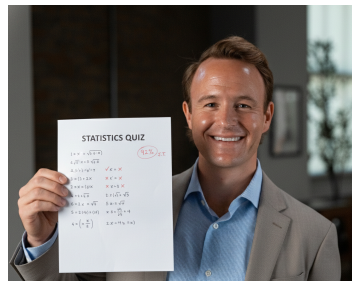


- Understanding the concepts really only comes from practice
- Class time will be mostly lecture and discussion. We will also have “in-class reps” on Quiz reviews and exercises.

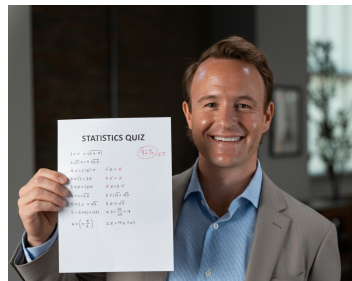
- 5 quizzes, taken during class on the 2nd, 4th, 6th, 8th, and 10th Friday



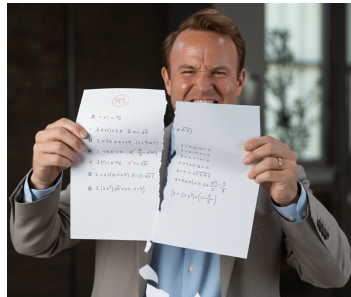
- 5 quizzes, taken during class on the 2nd, 4th, 6th, 8th, and 10th Friday
- Assessing your knowledge of lecture material up to that point



- 5 quizzes, taken during class on the 2nd, 4th, 6th, 8th, and 10th Friday
- Assessing your knowledge of lecture material up to that point
- Concepts assessed will come directly from the homeworks



- Why homework?



- [illegible]

- [illegible]

-



LLMs can be useful for the homeworks and learning concepts. I encourage their use! And ... I will not police their use. I only ask that you cite when LLMs are used.

However, if overused & relied on too heavily, **you will not learn anything.**

The quizzes are designed from the homeworks and will assess how well you understand the concepts. **Through the quizzes, you will get feedback quickly on whether you've overused AI!** Use this feedback to optimize your learning with AI.



- The exam will be during the 11th week (final exam week). There are no make-ups or alternative dates, so please plan accordingly!
- It will be a free response, written exam taken with pen and paper (exactly like our quizzes)



Homework	20%
Exam	30%
Quizzes	50%

- We will use **R** for statistical analysis throughout the course
- You are definitely welcome to use **Python**, but the course materials will (mostly) be in **R**
- This is industrial-strength, state-of-the-art, and free software for statistical computing
- We will access R through the **RStudio** graphical interface; make sure both are installed on your laptop and bring it to every class





- My office hours
- Schedule an one-on-one time with me to chat (Zoom or in-person)

Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff



- Simply put, the goal is to predict a target variable Y with input variables X
- In data mining terminology, this is known as supervised learning
- We observe training data: $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$



- Simply put, the goal is to predict a target variable Y with input variables X
- In data mining terminology, this is known as supervised learning
- We observe training data: $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$

Our goal: learn a function f such that $Y \approx f(X)$



- Simply put, the goal is to **predict a target variable Y** with **input variables X**
- In data mining terminology, this is known as **supervised learning**
- We observe training data: $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$

Our goal: learn a function f such that $Y \approx f(X)$

Tangent—what do you think **unsupervised learning** is?



In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$



In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$

- $f(X_i)$ is the systematic part (the signal)



In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$

- $f(X_i)$ is the systematic part (the signal)
- ϵ_i is the random error (the noise)



In general, a useful way to think about it is that Y and X are related in the following way:

$$Y_i = f(X_i) + \epsilon_i$$

- $f(X_i)$ is the systematic part (the signal)
- ϵ_i is the random error (the noise)
- The main purpose is to *learn or estimate* $f(\cdot)$ from data



1. **Regression:** When Y is quantitative (continuous)



1. **Regression:** When Y is quantitative (continuous)

- Examples: house prices, test scores, income, stock returns



1. **Regression:** When Y is quantitative (continuous)
 - Examples: house prices, test scores, income, stock returns
2. **Classification:** When Y is qualitative (categorical)



1. **Regression:** When Y is quantitative (continuous)
 - Examples: house prices, test scores, income, stock returns
2. **Classification:** When Y is qualitative (categorical)
 - Examples: win/lose, sick/healthy, pay/default, race, gender, ...



1. **Regression:** When Y is quantitative (continuous)
 - Examples: house prices, test scores, income, stock returns
2. **Classification:** When Y is qualitative (categorical)
 - Examples: win/lose, sick/healthy, pay/default, race, gender, ...



1. **Regression:** When Y is quantitative (continuous)
 - Examples: house prices, test scores, income, stock returns
2. **Classification:** When Y is qualitative (categorical)
 - Examples: win/lose, sick/healthy, pay/default, race, gender, ...

Today we'll focus on regression, but the same principles apply to classification



1. Choose a set of **training data**: $(Y_1, X_1), \dots, (Y_n, X_n)$

How do we estimate $f(\cdot)$?



1. Choose a set of **training data**: $(Y_1, X_1), \dots, (Y_n, X_n)$
2. Fit $f(\cdot)$ to training data using:



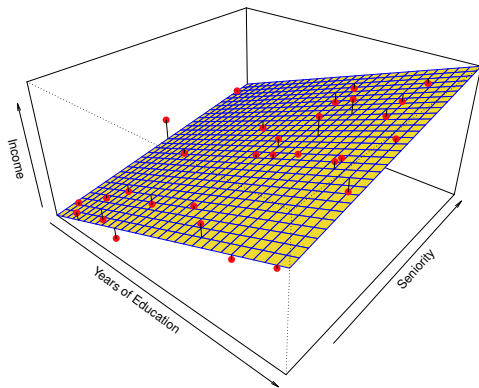
1. Choose a set of **training data**: $(Y_1, X_1), \dots, (Y_n, X_n)$
2. Fit $f(\cdot)$ to training data using:
 - **Parametric model**: Make assumptions about the form of f (e.g., linear)



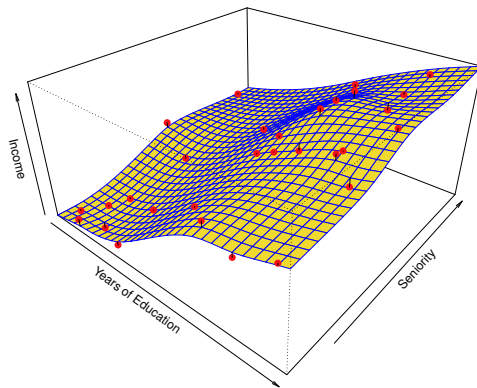
1. Choose a set of **training data**: $(Y_1, X_1), \dots, (Y_n, X_n)$
2. Fit $f(\cdot)$ to training data using:
 - **Parametric model**: Make assumptions about the form of f (e.g., linear)
 - **Nonparametric model**: Let the data determine the form of f (e.g., k-nearest neighbors)



1. Choose a set of **training data**: $(Y_1, X_1), \dots, (Y_n, X_n)$
2. Fit $f(\cdot)$ to training data using:
 - **Parametric model**: Make assumptions about the form of f (e.g., linear)
 - **Nonparametric model**: Let the data determine the form of f (e.g., k-nearest neighbors)
3. Evaluate performance on **testing data** and *adjust*



Parametric



Nonparametric

- **Parametric:** Restrictive assumptions, simple interpretation
- **Nonparametric:** Flexible assumptions, complex interpretation

Introduction to the course

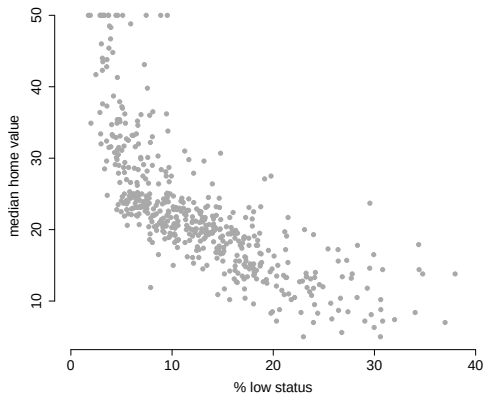
Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff

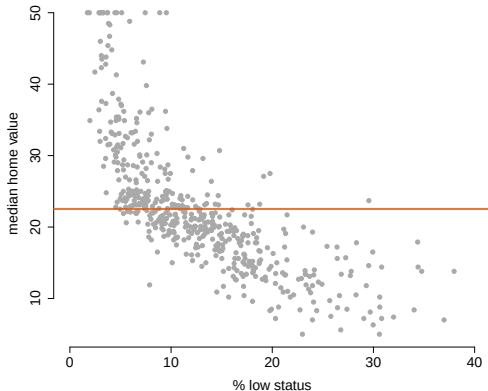
- Predict median home value with percent low economic status
- What should $f(\cdot)$ be?



What should $f(\cdot)$ be?

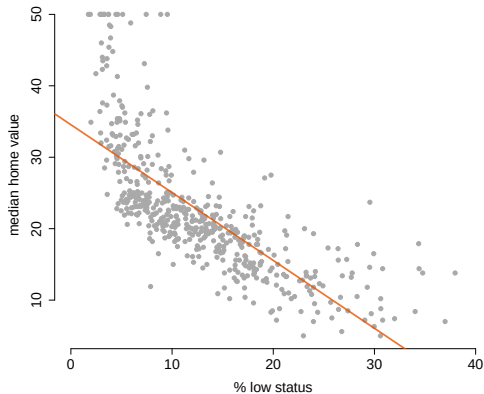


How about this? (Parametric: $Y = \mu + \epsilon$)



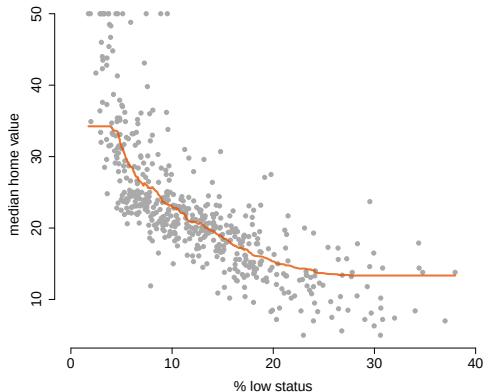
If % low status = 30, what is the prediction?

Linear model!



If % low status = 30, what is the prediction?

(k -NN with $k = 100$)



If % low status = 30, what is the prediction?



Prediction at point x : $\hat{f}(x)$ = average of k nearest points around x



Prediction at point x : $\hat{f}(x)$ = average of k nearest points around x

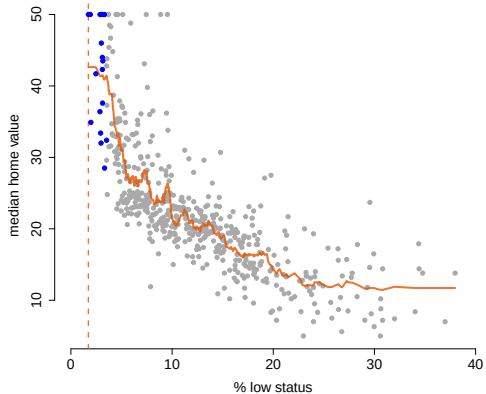
- Points that are **closer** to where we want to predict should be more relevant
- We choose the k points closest to the X value we're trying to predict
- This is the **k -nearest neighbors** algorithm

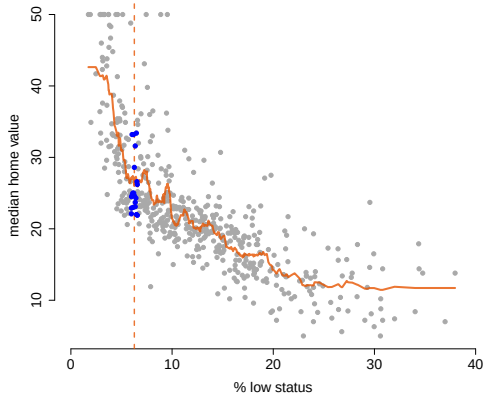


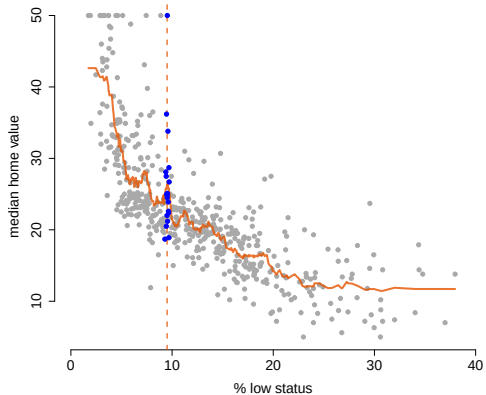
Prediction at point x : $\hat{f}(x)$ = average of k nearest points around x

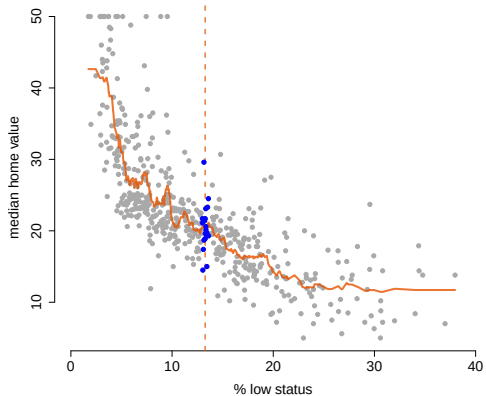
- Points that are **closer** to where we want to predict should be more relevant
- We choose the k points closest to the X value we're trying to predict
- This is the **k -nearest neighbors** algorithm

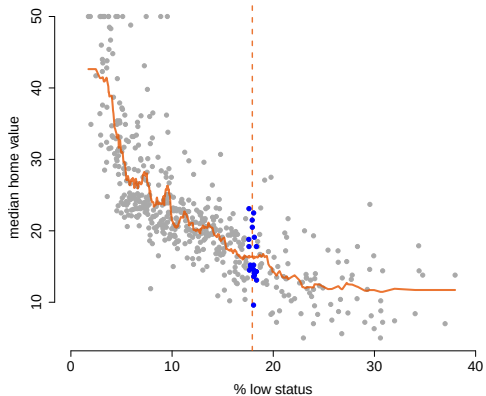
Let's see how this works with $k = 20$...

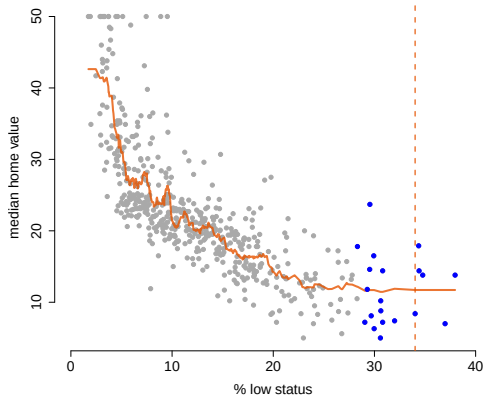




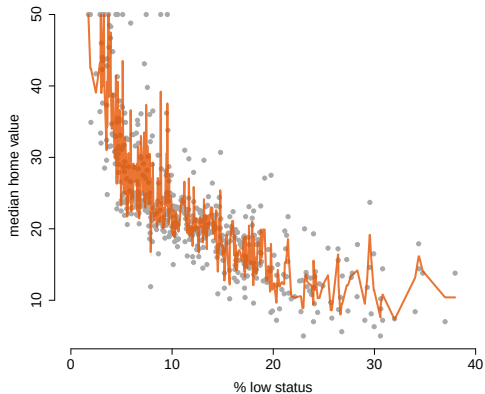




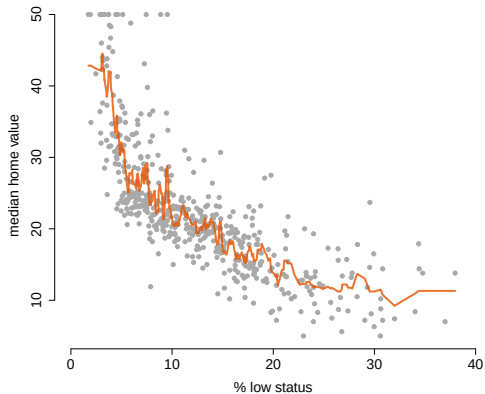




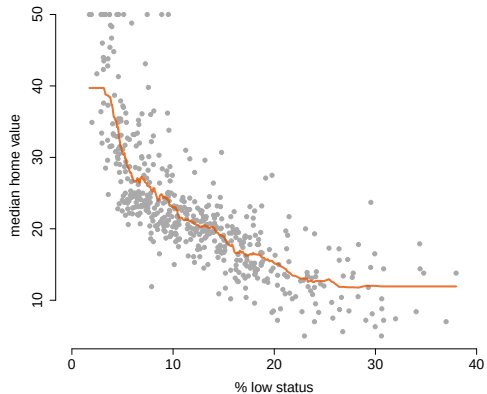
Why not choose $k = 2$?



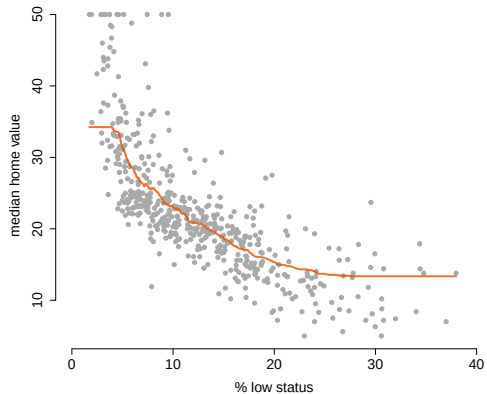
Or $k = 10$?



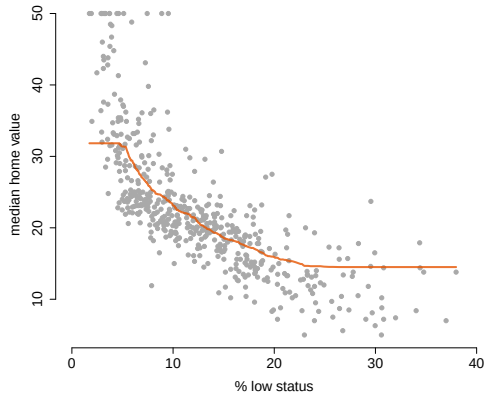
Or $k = 50$?



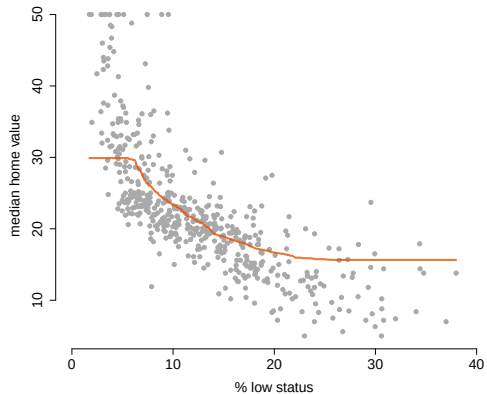
Or $k = 100$?



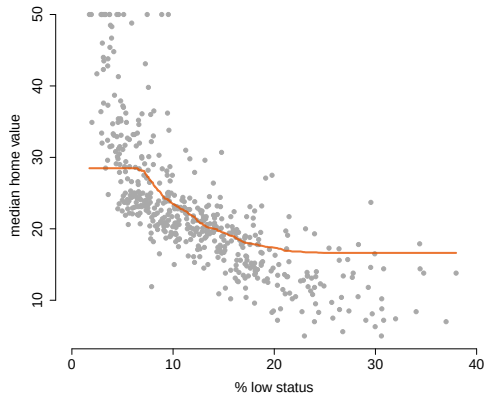
Or $k = 150$?



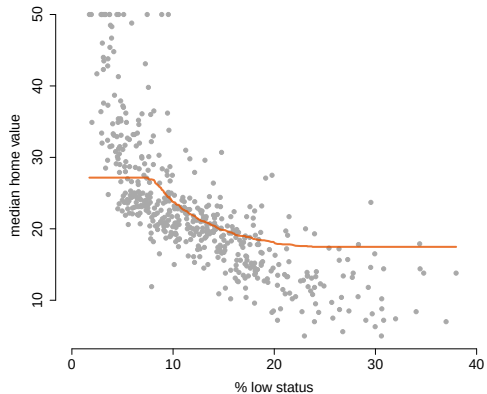
Or $k = 200$?



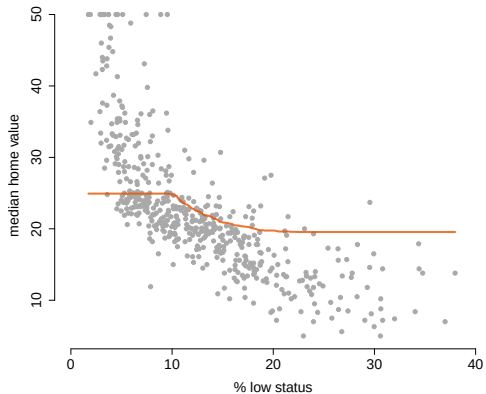
Or $k = 250$?



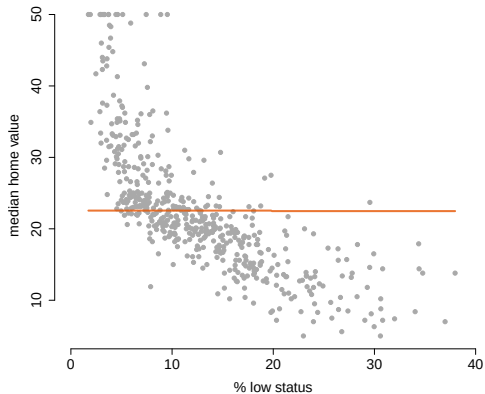
Or $k = 300$?



Or $k = 400$?



Or $k = 505$ (all data)?





- **In-sample error:** How well does the model fit the training data?

$$\text{RMSE}_{\text{in}} = \sqrt{\frac{1}{n} \sum_{i=1}^n [Y_i - \hat{f}(X_i)]^2}$$



- **In-sample error:** How well does the model fit the training data?

$$\text{RMSE}_{\text{in}} = \sqrt{\frac{1}{n} \sum_{i=1}^n [Y_i - \hat{f}(X_i)]^2}$$

- **Out-of-sample error:** How well does the model predict new data?

$$\text{RMSE}_{\text{out}} = \sqrt{\frac{1}{m} \sum_{i=1}^m [Y_i^o - \hat{f}(X_i^o)]^2}$$



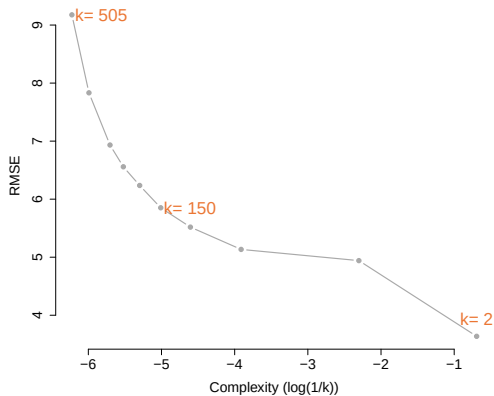
- **In-sample error:** How well does the model fit the training data?

$$\text{RMSE}_{\text{in}} = \sqrt{\frac{1}{n} \sum_{i=1}^n [Y_i - \hat{f}(X_i)]^2}$$

- **Out-of-sample error:** How well does the model predict new data?

$$\text{RMSE}_{\text{out}} = \sqrt{\frac{1}{m} \sum_{i=1}^m [Y_i^o - \hat{f}(X_i^o)]^2}$$

- We care about out-of-sample performance!



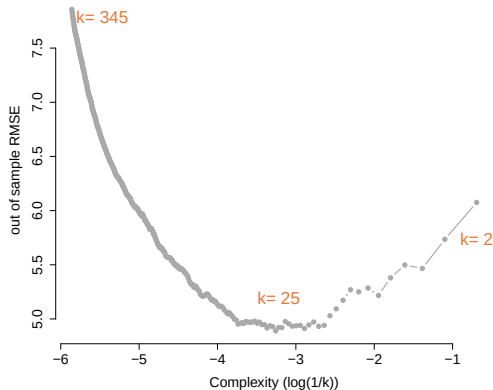
It looks like $k = 2$ is the best. Should we choose this model?



- Suppose we have m additional observations (X_i^o, Y_i^o) that we **did not use** to fit the model
- This is called the **validation set** (or hold-out set, or test set)
- We evaluate with out-of-sample RMSE:

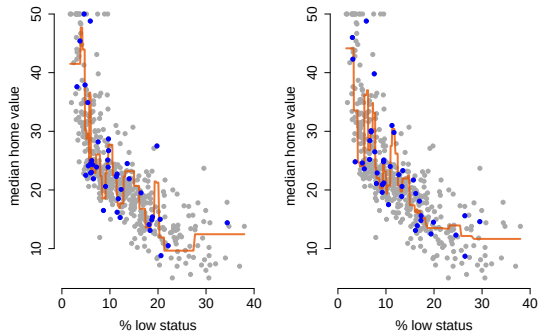
$$\text{RMSE}^o = \sqrt{\frac{1}{m} \sum_{i=1}^m [Y_i^o - \hat{f}(X_i^o)]^2}$$

Fit each model on training set (size 400). Test on validation set (size 106).



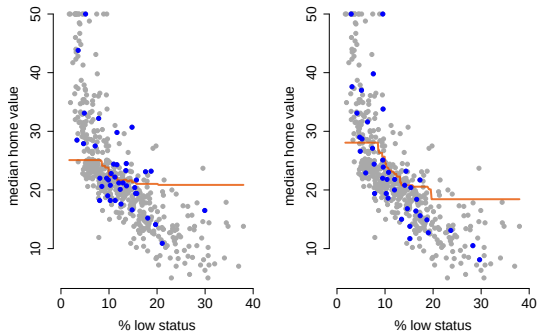
Now $k \approx 46$ looks like the best choice! Not too simple, but not too complex!

$k = 2$: Low bias, high variance



Training set of size 40

$k = 25$: High bias, low variance



Training set of size 40

Introduction to the course

Course logistics

Supervised learning

Building intuition: k-nearest neighbors

The bias-variance tradeoff



When fitting a predictive model, there is a tradeoff between:

- **Bias**: How far off our predictions are on average



When fitting a predictive model, there is a tradeoff between:

- **Bias**: How far off our predictions are on average
- **Variance**: How much our predictions vary across different training sets



When fitting a predictive model, there is a tradeoff between:

- **Bias**: How far off our predictions are on average
- **Variance**: How much our predictions vary across different training sets



When fitting a predictive model, there is a tradeoff between:

- **Bias**: How far off our predictions are on average
- **Variance**: How much our predictions vary across different training sets

Our goal: Find the sweet spot in the bias-variance tradeoff!



- **Simple models** (e.g., large k in k-NN):



- **Simple models** (e.g., large k in k-NN):
 - May not capture all patterns → **high bias**



- **Simple models** (e.g., large k in k-NN):
 - May not capture all patterns → **high bias**
 - But predictions are stable → **low variance**



- **Simple models** (e.g., large k in k-NN):
 - May not capture all patterns → **high bias**
 - But predictions are stable → **low variance**
- **Complex models** (e.g., small k in k-NN):



- **Simple models** (e.g., large k in k-NN):
 - May not capture all patterns → **high bias**
 - But predictions are stable → **low variance**
- **Complex models** (e.g., small k in k-NN):
 - Can capture complex patterns → **low bias**



- **Simple models** (e.g., large k in k-NN):
 - May not capture all patterns → **high bias**
 - But predictions are stable → **low variance**
- **Complex models** (e.g., small k in k-NN):
 - Can capture complex patterns → **low bias**
 - But predictions vary a lot → **high variance**



For a prediction at point x , the mean squared error can be decomposed:

$$\text{MSE} = \text{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\text{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Variance}}$$



For a prediction at point x , the mean squared error can be decomposed:

$$\text{MSE} = \text{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\text{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Variance}}$$

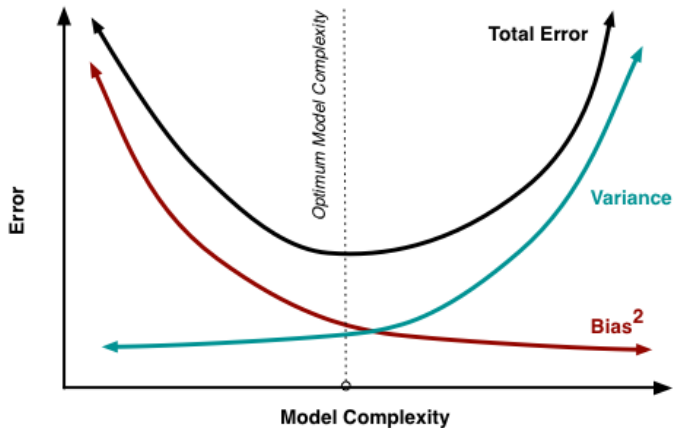
- **Bias²**: How far off our average prediction is from the truth



For a prediction at point x , the mean squared error can be decomposed:

$$\text{MSE} = \text{E}[(f(x) - \hat{f}(x))^2] = \underbrace{[\text{E}[\hat{f}(x)] - f(x)]^2}_{\text{Bias}^2} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Variance}}$$

- **Bias²**: How far off our average prediction is from the truth
- **Variance**: How much our predictions vary



- As model complexity increases, bias decreases but variance increases
- The optimal model complexity minimizes the total error



- A model that fits the training data perfectly may not generalize to new data



- A model that fits the training data perfectly may not generalize to new data
- This is called **overfitting**



- A model that fits the training data perfectly may not generalize to new data
- This is called **overfitting**
- We want models that:



- A model that fits the training data perfectly may not generalize to new data
- This is called **overfitting**
- We want models that:
 - Capture important patterns that generalize



- A model that fits the training data perfectly may not generalize to new data
- This is called **overfitting**
- We want models that:
 - Capture important patterns that generalize
 - Don't chase noise in the training data



- A model that fits the training data perfectly may not generalize to new data
- This is called **overfitting**
- We want models that:
 - Capture important patterns that generalize
 - Don't chase noise in the training data
- **The bias-variance tradeoff helps us understand this balance**



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods
3. There's a fundamental **bias-variance tradeoff**:



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods
3. There's a fundamental **bias-variance tradeoff**:
 - Simple models: high bias, low variance



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods
3. There's a fundamental **bias-variance tradeoff**:
 - Simple models: high bias, low variance
 - Complex models: low bias, high variance



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods
3. There's a fundamental **bias-variance tradeoff**:
 - Simple models: high bias, low variance
 - Complex models: low bias, high variance
4. We want to find the **sweet spot** that minimizes total prediction error



1. **Supervised learning** is about predicting Y from X by learning $f(\cdot)$
2. We can use **parametric** or **nonparametric** methods
3. There's a fundamental **bias-variance tradeoff**:
 - Simple models: high bias, low variance
 - Complex models: low bias, high variance
4. We want to find the **sweet spot** that minimizes total prediction error
5. **Out-of-sample performance** is what matters, not in-sample fit



- Up next, we'll start **inference for regression** and **multiple regression**



- Up next, we'll start **inference for regression** and **multiple regression**
- We'll learn how to:



- Up next, we'll start **inference for regression** and **multiple regression**
- We'll learn how to:
 - Fit and interpret linear regression models



- Up next, we'll start **inference for regression** and **multiple regression**
- We'll learn how to:
 - Fit and interpret linear regression models
 - Make statistical inferences about regression coefficients



- Up next, we'll start **inference for regression** and **multiple regression**
- We'll learn how to:
 - Fit and interpret linear regression models
 - Make statistical inferences about regression coefficients
 - Extend simple regression to multiple predictors



- Up next, we'll start **inference for regression** and **multiple regression**
- We'll learn how to:
 - Fit and interpret linear regression models
 - Make statistical inferences about regression coefficients
 - Extend simple regression to multiple predictors
 - Assess model fit and make predictions



- Up next, we'll start **inference for regression** and **multiple regression**
- We'll learn how to:
 - Fit and interpret linear regression models
 - Make statistical inferences about regression coefficients
 - Extend simple regression to multiple predictors
 - Assess model fit and make predictions
- The bias-variance tradeoff will continue to guide our thinking throughout the course!

Extra Slides



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$
2. **Variance:** Our specific prediction $\hat{f}(X)$ varies around its average



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$
2. **Variance:** Our specific prediction $\hat{f}(X)$ varies around its average
3. **Noise:** The true Y has random error ϵ even if we knew $f(X)$ perfectly



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$
2. **Variance:** Our specific prediction $\hat{f}(X)$ varies around its average
3. **Noise:** The true Y has random error ϵ even if we knew $f(X)$ perfectly



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$
2. **Variance:** Our specific prediction $\hat{f}(X)$ varies around its average
3. **Noise:** The true Y has random error ϵ even if we knew $f(X)$ perfectly

Start with $Y = f(X) + \epsilon$. Decompose the error by adding/subtracting $E[\hat{f}(X)]$:

$$Y - \hat{f}(X) = [f(X) - E[\hat{f}(X)]] + [E[\hat{f}(X)] - \hat{f}(X)] + \epsilon$$



The prediction error $Y - \hat{f}(X)$ has three sources:

1. **Bias:** Our model's average prediction $E[\hat{f}(X)]$ might be off from the true $f(X)$
2. **Variance:** Our specific prediction $\hat{f}(X)$ varies around its average
3. **Noise:** The true Y has random error ϵ even if we knew $f(X)$ perfectly

Start with $Y = f(X) + \epsilon$. Decompose the error by adding/subtracting $E[\hat{f}(X)]$:

$$Y - \hat{f}(X) = [f(X) - E[\hat{f}(X)]] + [E[\hat{f}(X)] - \hat{f}(X)] + \epsilon$$

Squaring both sides gives squared terms **plus cross terms**:

$$\begin{aligned}(Y - \hat{f}(X))^2 &= [f(X) - E[\hat{f}(X)]]^2 + [E[\hat{f}(X)] - \hat{f}(X)]^2 + \epsilon^2 \\ &\quad + (\text{cross terms})\end{aligned}$$



Key insight: When we take expectations, the cross terms vanish!



Key insight: When we take expectations, the cross terms vanish!

- $E[E[\hat{f}(X)] - \hat{f}(X)] = 0$ (centered around its mean)



Key insight: When we take expectations, the cross terms vanish!

- $E[E[\hat{f}(X)] - \hat{f}(X)] = 0$ (centered around its mean)
- $E[\epsilon] = 0$ (noise has mean zero)



Key insight: When we take expectations, the cross terms vanish!

- $E[E[\hat{f}(X)] - \hat{f}(X)] = 0$ (centered around its mean)
- $E[\epsilon] = 0$ (noise has mean zero)
- ϵ is independent of $\hat{f}(X)$ (noise is independent of our model)



Key insight: When we take expectations, the cross terms vanish!

- $E[E[\hat{f}(X)] - \hat{f}(X)] = 0$ (centered around its mean)
- $E[\epsilon] = 0$ (noise has mean zero)
- ϵ is independent of $\hat{f}(X)$ (noise is independent of our model)



Key insight: When we take expectations, the cross terms vanish!

- $E[E[\hat{f}(X)] - \hat{f}(X)] = 0$ (centered around its mean)
- $E[\epsilon] = 0$ (noise has mean zero)
- ϵ is independent of $\hat{f}(X)$ (noise is independent of our model)

For the MSE $E[(f(X) - \hat{f}(X))^2]$, the cross terms vanish, leaving only the two squared terms: bias^2 and variance.